

Ankit Shukla

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Senior Software Engineer - C/C++, Embedded & Systems Programming

Professional Summary

Software engineer with 14 years of experience building and shipping systems-level C/C++ software on Linux, Windows, QNX, and WinCE across storage, automotive, avionics, and graphics domains. Strong background in multi-threaded and distributed systems, IPC, socket programming, and Linux internals, with hands-on experience taking features from design through implementation, unit testing, automation, and production on-call support. Comfortable working across the stack from device drivers and kernel-adjacent code to cloud-integrated services on AWS and Azure. Experience leading a team of six and mentoring engineers through code review and design discussions.

Core Technical Skills

Languages: C, C++ (11/14/17, STL, multithreading), Python, Go, Java

Systems: Linux systems programming, kernel internals & device drivers, Linux graphics stack, multithreading, IPC & synchronization, socket programming, TCP/IP

Platforms: Linux (Ubuntu, Kali, OpenSUSE), Windows, WinCE 7, QNX, LynxOS, AIX, Solaris | x86, ARM

Design & Architecture: SOLID principles, Design Patterns (Singleton, Factory, Adapter, Builder), Data Structures & Algorithms, REST/SOAP/gRPC, JSON/XML

Cloud & Security: AWS, Azure, OpenSSL 3.0 (RSA signature verification, AES-CBC encryption)

Tooling: Visual Studio 2019/2022, QNX Momentics, Qt, Eclipse, PyCharm, IntelliJ, VS Code, GDB, Trace32, Lauterbach

Build & CI/CD: Make, CMake, Bazel, Apache Maven, MS Build, Jenkins

Testing & QA: GoogleTest/GoogleMock, VectorCast, Ginkgo, TestRail

Version Control & PM: Git, SVN, RTC, Bitbucket, Gerrit, Jira, IBM Rational DOORS

AI-assisted development: Cursor and GitHub Copilot with Claude models

Professional Experience

Cohesity Inc.

Dec 2024 - Present

Software Engineer - Resource Manager (Heimdall) | Storage

- Built core components of Heimdall, an internal resource manager that provisions and tracks Linux/Windows VM fleets and disks used by backup services, in C++, Python, and Go.
- Owned features end-to-end: wrote design docs, drove requirement discussions with dependent teams, broke work into epics/stories in Jira, and implemented and code-reviewed the resulting changes.
- Wrote unit tests (GoogleTest/GoogleMock) and integrated automated test runs through Ginkgo, publishing results to TestRail for QA sign-off.
- Investigated and resolved release-blocking defects, including issues surfaced during on-call rotations for customer-impacting backup failures.
- Maintained the build/image scripts used to produce Heimdall deployment images across environments.

Veritas Technologies

Jan 2023 - Dec 2024

Software Engineer - Shared Licensing Component (SLIC) | Storage

- Developed licensing/entitlement features in C++ for SLIC, the shared license-validation component used across Veritas's product line, built on OpenSSL and Poco.
- Designed and implemented on Windows, then ported and validated changes across Red Hat, OpenSUSE, AIX, and Solaris.

- Rebuilt the Poco and OpenSSL dependencies on each supported platform and delivered updated SLIC SDKs to downstream product teams to close known CVEs.
- Triaged vulnerabilities flagged by Black Duck in OpenSSL/Poco and assessed real-world impact on SLIC before prioritizing fixes.

Siemens Mobility

Mar 2019 - Jan 2023

Senior Software Engineer / Technical Lead - Radio Block Center (RBC) | Rail / Mobility

- Led a team of 6 engineers on RBC, safety-critical software that receives train position reports over Balise devices and issues movement authority to trains.
- Delivered new features in C/C++/Python from requirements through implementation, static analysis, and unit testing with VectorCast.
- Ran code reviews and diagnosed field issues from logs, driving root-cause analysis and fixes for the team.
- Built two internal automation tools in Python (Tkinter): one to automate environment/software setup for the project, another to update code baselines and timestamps across unit-test suites — both adopted team-wide to cut manual setup time.
- Owned monthly billing releases and forecast updates for the engagement.

Tata Elxsi Ltd

Jun 2015 - Mar 2019

Software Engineer - Automotive Infotainment & Instrument Cluster | Automotive

- Applink (Panasonic Automotive, 3 yrs 2 mo): built C/C++ modules for a smartphone-to-infotainment connectivity system on QNX/Windows, covering voice command handling, touch UI input, and embedded navigation; authored design docs, analyzed core dumps, and fixed defects found in production logs.
- GM Instrument Cluster (Visteon, 6 mo): processed vehicle bus data and drove UI widgets showing tire pressure, fuel level, and speed/tachometer readings on QNX/Windows; delivered features from system design through GoogleTest/GoogleMock unit testing and static analysis.
- Used Singleton and related design patterns to keep shared hardware-access code consistent across both projects.

Infosys Ltd

Aug 2014 - Jun 2015

Software Engineer - Graphics Driver Development (AMD) | Graphics / Multimedia

- Ported the Linux OpenGL/LLVM call stack to WinCE and worked on the Radeon GPU driver's DRM subsystem to bring hardware-accelerated rendering to WinCE-based boards (Ontario, Kabini, Mullins).
- Diagnosed and fixed driver defects affecting video and gaming performance, and built OpenGL test applications to validate rendering correctness.

Rockwell Collins India Pvt. Ltd

Aug 2011 - Aug 2014

Software Engineer - Onboard Maintenance System | Avionics

- Developed features in C/C++ for OMS, an avionics subsystem (Diagnostic Report and Display Manager applications) that communicates with aircraft LRUs over ARINC/Ethernet and XML-RPC to surface maintenance data to flight crews.
- Worked on a 20-person program using Trace32/Lauterbach for hardware-level debugging on LynxOS targets, with requirements tracked in IBM Rational DOORS.
- Wrote and executed functional and unit tests and fixed defects found during integration testing.

Education

B.Tech: UPTU — 2010

Diploma in System Software: Pune University — 2011